Cardiovascular System (CARD) · NPLEX-P1-CARD-04

CV-4 Heart Valves and the Cardiac Cycle

Brain dumps can be right in more than one way. Grade the ideas and the relationships, not the wording. A response is competent when it carries the essentials below.

NPLEX-P1-CARD-04-01-C01-BD1

Brain dump 1 · foundational retrieval: Draw the four valves and where each is heard

A competent response contains

· Tricuspid: right atrium to right ventricle, anterior, posterior, and septal cusps, held closed by chordae running to three papillary muscles, heard at the left lower sternal border in the 4th to 5th intercostal space.

· Mitral: left atrium to left ventricle, anterior and posterior cusps, held closed by chordae to the anterolateral and posteromedial papillary muscles, heard at the apex in the left 5th intercostal space at the midclavicular line.

· Pulmonary: right ventricle to pulmonary trunk, anterior, right, and left cusps, sealed by cusp shape with no chordae, heard at the left 2nd intercostal space at the sternal border.

· Aortic: left ventricle to aorta, right, left, and posterior cusps also called right coronary, left coronary, and noncoronary, sealed by cusp shape with no chordae, heard at the right 2nd intercostal space at the sternal border.

· The fibrous skeleton is dense connective tissue rings that hold the four valve openings in shape and electrically insulate the atria from the ventricles.

· Each cusp is a fold of endocardium with a connective tissue core and no muscle, so the valve has no power of its own and moves only as the pressure gradient across it changes: open when the pressure behind exceeds the pressure in front, closed when that reverses.

· Semilunar cusps seal because backflow at the end of systole fills the pockets and presses them together, with a central nodule and thin lateral lunules completing the seal.

· The coronary arteries arise from the right and left aortic sinuses behind the cusps, which is where the cusp names come from.

· Aortic stenosis radiates to the carotids because aortic sounds follow the ascending aorta up to the right 2nd intercostal space.

· Mitral regurgitation radiates to the left axilla along the direction of the backward jet.

· A bicuspid aortic valve occurs in about 1% to 2% of people and calcifies and becomes stenotic early.

Distinctions that matter

· AV valves need chordae because their large thin cusps face full systolic pressure; semilunar valves need none because their pocket shape does the sealing.

· The valves all sit close together behind the sternum, so an auscultation area is where the sound is carried downstream or over the nearest chamber rather than where the valve lies.

Common misconceptions

· Writing that papillary muscles pull the AV valves open in diastole. They contract with the ventricle and hold the cusps shut in systole.

· Treating an auscultation area as the surface marking of the valve itself. The areas follow the path of the sound, which is why aortic sounds are loudest to the right of the sternum.

Where it is taught: Section C01 of the CV-4 Heart Valves and the Cardiac Cycle notes.

NPLEX-P1-CARD-04-01-C01-BD2

Brain dump 2 · integration and application: Why an inferior infarct tears one papillary muscle and not the other

A competent response contains

· The posteromedial papillary muscle failed, because it usually depends on the posterior descending artery alone, which arises from the right coronary artery in most hearts.

· The anterolateral papillary muscle usually has a dual supply from left anterior descending and circumflex branches, so it is far less likely to infarct and rupture.

· Sequence: occlusion of the right coronary or posterior descending artery, necrosis of the posteromedial muscle, rupture of the softened muscle usually within the first week, mitral cusps flail into the left atrium during systole, acute severe mitral regurgitation into a small noncompliant atrium, large v waves with pulmonary edema and low forward stroke volume.

· The regurgitant volume goes backward into the left atrium from the onset of contraction, so left atrial and pulmonary venous pressures rise steeply.

· Acute regurgitation is tolerated badly because the atrium has had no time to dilate, so a given regurgitant volume produces a much higher pressure than in chronic disease, where a dilated compliant atrium absorbs it.

· In chronic mitral regurgitation the ventricle also has time to undergo eccentric hypertrophy and dilate, which protects forward output for years.

· Chordal rupture from myxomatous degeneration and from endocarditis produces the same loss of cusp support.

Distinctions that matter

· Which papillary muscle ruptures is decided by blood supply, not by size or position.

· Severity of the regurgitation and tolerance of it are separate questions: compliance of the receiving atrium decides the pressure, and time decides the compliance.

Common misconceptions

· Expecting a loud murmur to mean a large leak. The murmur may be short or soft when the atrium is small and stiff and pressures equalise quickly.

· Assuming total stroke volume must fall. The ventricle ejects a large total volume, but much of it goes backward, so forward output drops.

Where it is taught: Section C01, papillary muscle paragraphs and the rupture flow diagram of the CV-4 Heart Valves and the Cardiac Cycle notes.

NPLEX-P1-CARD-04-01-C01-CA01

Clinical application: Sudden breathlessness five days after an inferior infarction

A competent response contains

· Recognize: acute severe mitral regurgitation from rupture of the posteromedial papillary muscle, following right coronary or posterior descending artery occlusion in an inferior infarction.

· Explain: the posteromedial muscle depends on the posterior descending artery alone, so it necroses and then ruptures in the first week; the cusps flail into the left atrium during systole, giving a holosystolic murmur at the apex radiating to the axilla, a large late systolic v wave on the wedge tracing, and a high mean wedge pressure that drives pulmonary edema and crackles.

· Differentiate: ventricular septal rupture gives a harsh holosystolic murmur at the left lower sternal border with a left to right shunt rather than a giant v wave, and free wall rupture with tamponade gives hypotension with a raised venous pressure, a prominent x descent, and no new loud murmur.

· Investigate: on the wedge tracing look for a small a wave with a very large late systolic v wave and a raised mean pressure; on the jugular tracing look at the height of the pulsation and the wave pattern, which here reflects the raised filling pressures rather than tricuspid regurgitation.

· Decide: blood pressure is low because a large share of each ejected volume goes backward into the left atrium, so forward stroke volume into the aorta falls even though total stroke volume has risen.

· Justify: the leak begins as soon as ventricular pressure exceeds atrial pressure, well before the aortic valve opens, and the atrium has not dilated, so it is stiff and its pressure rises steeply, which both limits further backward flow into a high pressure chamber and transmits pressure to the pulmonary veins; the tachycardia is the compensation for the small forward volume.

Distinctions that matter

· A new murmur after infarction is placed by its location and timing: apex radiating to the axilla points to the mitral valve, left lower sternal border points to the septum.

· A giant v wave reports atrial compliance and the volume arriving backward, not the size of the valve opening.

Common misconceptions

· Reading the normal electrocardiogram as evidence against a mechanical complication. Rupture is a structural event and does not change the tracing.

· Assuming low blood pressure means poor contractility. Contractility can be preserved while forward output falls because the ventricle has a second low pressure outlet.

Where it is taught: Section C01, papillary muscle paragraphs and rupture flow diagram, with the large v wave in section C04 of the CV-4 Heart Valves and the Cardiac Cycle notes.

NPLEX-P1-CARD-04-01-C02-BD1

Brain dump 1 · foundational retrieval: Draw the Wiggers diagram and mark every valve event

A competent response contains

· Four valve events: mitral closure when left ventricular pressure exceeds left atrial pressure, aortic opening when ventricular pressure exceeds aortic diastolic pressure of about 80 mm Hg, aortic closure when flow reverses as ventricular pressure falls below aortic pressure, and mitral opening when ventricular pressure falls below left atrial pressure.

· Phases in order: atrial systole, isovolumetric contraction of about 0.05 s, rapid ejection, reduced ejection, isovolumetric relaxation, rapid filling, diastasis.

· Normal numbers: aortic diastolic pressure about 80 mm Hg, peak ventricular and aortic pressure about 120 mm Hg, end-diastolic volume about 120 mL, end-systolic volume about 50 mL, stroke volume about 70 mL, ejection fraction about 55% to 70%, pulmonary artery pressure about 25/8 mm Hg.

· At about 75 beats per minute the cycle lasts about 0.8 s, with systole about 0.3 s and diastole about 0.5 s.

· About two thirds of the stroke volume leaves during rapid ejection; in reduced ejection the T wave begins and ventricular pressure falls slightly below aortic pressure while momentum carries blood forward.

· S1 is mitral then tricuspid closure, best at the apex, from vibration of the tensed cusps, chordae, walls, and blood; aortic valve opening is silent unless a stiff domed valve gives an ejection click.

· S2 is aortic then pulmonic closure, best at the base, and the dicrotic notch appears on the aortic curve at that moment from the backflow and aortic rebound.

· Atrial contraction after the P wave supplies about the last 20% of end-diastolic volume at rest.

· Rapid filling delivers most of the filling and is aided by active relaxation; diastasis is the reduced filling period that follows.

Distinctions that matter

· Every valve event on the diagram is a crossing of two pressure curves, so each one is read by asking which pressure across that valve is higher.

· Only the two isovolumetric phases have all four valves closed, which is why volume is flat in both.

Common misconceptions

· Drawing aortic pressure falling to zero in diastole. It falls to about 80 mm Hg, which is the pressure the ventricle has to exceed to open the valve.

· Marking S1 and S2 as valve opening sounds. Both are closure sounds, and normal opening is silent.

Where it is taught: Section C02, systole and diastole paragraphs and the cardiac cycle diagram of the CV-4 Heart Valves and the Cardiac Cycle notes.

NPLEX-P1-CARD-04-01-C02-BD2

Brain dump 2 · integration and application: Name the phase from four numbers, and what a fast rate costs

A competent response contains

· Tracing A: ventricular pressure above atrial pressure so the mitral valve is closed, aortic pressure above ventricular pressure so the aortic valve is closed, and a volume of 120 mL is end-diastolic, so this is isovolumetric contraction just after S1.

· Tracing B: the same pressure relationships with a volume of 50 mL, which is end-systolic, so this is isovolumetric relaxation just after S2.

· Volume, not pressure, separates the two isovolumetric phases, because both have all valves closed and the same pressures can occur on the way up and on the way down.

· Tracing C: atrial pressure above ventricular pressure so the mitral valve is open, aortic pressure well above ventricular pressure so the aortic valve is closed, and a volume between end-systolic and end-diastolic, so this is ventricular filling.

· Raising the rate to 160 beats per minute shortens the whole cycle, but diastole shortens far more than systole.

· Diastasis, the reduced filling period, is lost first, so filling depends more on rapid filling and on atrial contraction, whose share of end-diastolic volume rises.

· Left ventricular coronary flow occurs mainly in diastole, so a short diastole reduces perfusion at the same time as demand is higher.

· The pulmonary valve opens before the aortic valve because the right ventricle has to exceed only about 8 mm Hg of pulmonary diastolic pressure, and it closes after the aortic valve because the low impedance pulmonary circulation prolongs forward flow, so A2 precedes P2.

Distinctions that matter

· Comparing two pressures at a time settles the state of each valve; trying to read the phase from one pressure alone does not.

· Shorter diastole reduces both filling and coronary perfusion, which are separate consequences of the same change.

Common misconceptions

· Assuming systole and diastole shorten in proportion as the rate rises. Diastole takes almost all of the loss.

· Reading a high ventricular pressure as proof the aortic valve is open. It is open only while ventricular pressure exceeds aortic pressure.

Where it is taught: Section C02, opening paragraph, right heart timing paragraph, and the worked example on identifying the phase of the CV-4 Heart Valves and the Cardiac Cycle notes.

NPLEX-P1-CARD-04-01-C02-RT01

Retrieval exercise: Rebuild the valve events of one left-sided cycle

A competent response contains

· Step 1: atrial systole completes filling to end-diastolic volume.

· Step 2: left ventricular pressure exceeds left atrial pressure and the mitral valve closes, producing S1.

· Step 3: isovolumetric contraction, all valves closed and pressure rising at constant volume.

· Step 4: ventricular pressure exceeds aortic pressure at about 80 mm Hg and the aortic valve opens.

· Step 5: rapid then reduced ejection, volume falling to end-systolic volume.

· Step 6: flow reverses and the aortic valve closes, producing S2 and the dicrotic notch.

· Step 7: isovolumetric relaxation, all valves closed and pressure falling at constant volume.

· Step 8: ventricular pressure falls below left atrial pressure and the mitral valve opens.

· Step 9: rapid filling, with an S3 if one is present, then diastasis.

· The dicrotic notch is the small downward deflection and rebound on the aortic curve produced by the brief backflow and the elastic recoil of the aorta at the moment the valve closes.

· S1 is loud with a short PR interval or mitral stenosis, because the cusps are still widely open when contraction begins and travel further to close, and soft with a long PR interval, because the cusps have already drifted together.

Distinctions that matter

· Steps 3 and 7 are the only all valves closed steps, and the volume is flat in both; what differs is whether the ventricle holds end-diastolic or end-systolic volume.

· Pressure thresholds belong to the opening steps and pressure crossings to the closing steps, so name the two pressures each time rather than a single number.

Common misconceptions

· Putting the aortic valve opening at the same moment as S1. Isovolumetric contraction sits between them.

· Explaining a loud S1 by stronger contraction alone. Cusp position at the start of contraction is what the notes make the deciding factor.

Where it is taught: Section C02, the cardiac cycle diagram and the paragraph on S1 intensity of the CV-4 Heart Valves and the Cardiac Cycle notes.

NPLEX-P1-CARD-04-01-C03-BD1

Brain dump 1 · foundational retrieval: Draw the normal pressure-volume loop

A competent response contains

· The loop is traced counterclockwise.

· Bottom right corner: mitral valve closure, at about 120 mL and about 8 mm Hg.

· Top right corner: aortic valve opening, at about 80 mm Hg.

· Top left corner: aortic valve closure, at about 50 mL and about 100 mm Hg.

· Bottom left corner: mitral valve opening.

· Right side is vertical isovolumetric contraction, the top is ejection peaking near 120 mm Hg, the left side is vertical isovolumetric relaxation, and the bottom edge is filling with little rise in pressure in a compliant ventricle.

· The width of the loop, end-diastolic volume minus end-systolic volume, is the stroke volume, and the enclosed area is the stroke work.

· The bottom edge follows the end-diastolic pressure-volume relationship, which is steeper in a stiff ventricle.

· The top left corner lies on the end-systolic pressure-volume line, whose slope rises with contractility.

· Normal values: end-diastolic volume about 120 mL, end-systolic volume about 50 mL, stroke volume about 70 mL, ejection fraction about 55% to 70%.

Distinctions that matter

· Width measures stroke volume and area measures stroke work, so a tall narrow loop can do more work while moving less blood.

· The diastolic edge reports compliance and the end-systolic corner reports contractility; they are different properties on different parts of the loop.

Common misconceptions

· Tracing the loop clockwise. Filling runs left to right along the bottom and ejection runs right to left along the top, which makes it counterclockwise.

· Reading the bottom edge as passive and featureless. Its slope is the stiffness of the ventricle and it rises steeply in a hypertrophied one.

Where it is taught: Section C03, first two paragraphs of the CV-4 Heart Valves and the Cardiac Cycle notes.

NPLEX-P1-CARD-04-01-C03-BD2

Brain dump 2 · integration and application: Name the lesion from the loop

A competent response contains

· Loop 1: stroke volume 125 mL, a volume load, aortic regurgitation; the giveaway is that the left side slopes to the right, so the ventricle gains volume while pressure falls after ejection, which means it is filling from the aorta.

· Loop 2: stroke volume 50 mL, a pressure load, aortic stenosis; the giveaway is a peak ventricular pressure of 215 mm Hg against a normal aortic pressure, which is a gradient across the valve.

· Loop 3: stroke volume 130 mL, a volume load, mitral regurgitation; the giveaway is that volume starts falling at the bottom right corner while pressure is only about 15 mm Hg, far too low to have opened the aortic valve.

· In aortic regurgitation relaxation is no longer isovolumetric because blood leaks back from the aorta as soon as ventricular pressure falls, so volume rises during it; chronic eccentric hypertrophy follows, the ventricle becomes more compliant and the loop shifts right, pulse pressure widens, and the large total stroke volume comes from filling by both atrium and aorta with the Frank-Starling mechanism.

· In mitral regurgitation contraction is no longer isovolumetric because blood escapes into the left atrium as soon as ventricular pressure exceeds atrial pressure, so volume falls during it; eccentric hypertrophy with left atrial dilation follows, total stroke volume rises but forward output falls, and end-systolic volume tends to fall early because the low pressure atrium is a second outlet.

· In aortic stenosis both isovolumetric phases are preserved; concentric hypertrophy follows, which normalises wall stress but stiffens the ventricle, steepening the diastolic edge and raising end-diastolic pressure, and the extra loop area means greater stroke work and oxygen demand.

· Peak pressures of about 180 to 250 mm Hg are quoted for severe aortic stenosis while aortic pressure stays normal or low, and incomplete ejection raises end-systolic volume and narrows the loop.

· Mitral stenosis underfills the ventricle, so the loop is small and shifted left.

Distinctions that matter

· A regurgitant lesion abolishes the isovolumetric phase in which the leaking valve ought to be closed, which is the quickest way to tell the two regurgitations apart on a loop.

· Total and forward stroke volume separate in mitral regurgitation, so loop width alone overstates what reaches the aorta.

Common misconceptions

· Taking a wide loop as a healthy one. In mitral regurgitation much of that width is volume going backward.

· Expecting raised peak pressure in every valve lesion. Peak pressure is raised in aortic stenosis and normal or low in mitral regurgitation, which is what rules stenosis out.

Where it is taught: Section C03, valve lesion paragraphs, the loop comparison table, and the worked example on reading a lesion of the CV-4 Heart Valves and the Cardiac Cycle notes.

NPLEX-P1-CARD-04-01-C03-CP01

Comparison table: Pressure-volume loop in three valve lesions

A competent response contains

· Type of load: aortic stenosis is a pressure load on afterload, aortic regurgitation and mitral regurgitation are volume loads.

· Peak left ventricular pressure: greatly increased in aortic stenosis with a gradient from ventricle to aorta, increased in aortic regurgitation, normal or decreased in mitral regurgitation.

· End-diastolic volume: normal or mildly increased in aortic stenosis, greatly increased in aortic regurgitation, increased in mitral regurgitation.

· End-systolic volume: increased in aortic stenosis, increased in aortic regurgitation, normal or decreased early in mitral regurgitation.

· Total stroke volume: normal or decreased in aortic stenosis, greatly increased in aortic regurgitation, increased in mitral regurgitation with forward volume reduced.

· Phase that is not truly isovolumetric: none in aortic stenosis, relaxation in aortic regurgitation with volume rising, contraction in mitral regurgitation with volume falling.

· Chronic hypertrophy: concentric in aortic stenosis, eccentric in aortic regurgitation, eccentric with left atrial dilation in mitral regurgitation.

· Loop shape: aortic stenosis is tall and narrow with a steep diastolic edge, aortic regurgitation is wide and shifted right with a sloping left side, mitral regurgitation is wide and shifted right with a sloping right side.

Distinctions that matter

· The sloping side names the lesion: a sloping left side is aortic regurgitation and a sloping right side is mitral regurgitation.

· Pressure loading gives concentric hypertrophy with sarcomeres added in parallel, volume loading gives eccentric hypertrophy with sarcomeres added in series.

Common misconceptions

· Filling the end-systolic volume row for mitral regurgitation as increased. Early on it tends to fall, because the atrium is a second low pressure outlet.

· Writing a raised peak pressure for every lesion that enlarges the loop. Only aortic stenosis raises it greatly, and it does so against a normal aortic pressure.

Where it is taught: Section C03, valve lesion paragraphs and the loop comparison table of the CV-4 Heart Valves and the Cardiac Cycle notes.

NPLEX-P1-CARD-04-01-C03-Q01

Board-style question: A loop that loses volume before the aortic valve opens

Correct answer: C. Blood escaping into the left atrium from the onset of ventricular contraction

Why it is right: Mitral regurgitation lets blood escape into the low pressure left atrium as soon as ventricular pressure exceeds atrial pressure, so volume falls during what should be isovolumetric contraction, the right side of the loop slopes, total stroke volume rises, and forward output falls.

A is wrong: A fixed outflow obstruction would raise peak ventricular pressure well above aortic pressure and give a tall narrow loop with both isovolumetric phases intact, but the peak here is 108 mm Hg.

B is wrong: Reduced compliance steepens the bottom edge of the loop and raises end-diastolic pressure without letting volume leave before the aortic valve opens, and the end-diastolic volume here is large rather than small.

D is wrong: Blood returning from the aorta would add volume during relaxation and make the left side of the loop slope to the right, which is the opposite side and the opposite direction of volume change.

A competent response contains

· Stroke volume is 125 mL, a volume load, with a normal peak ventricular pressure.

· Volume falling at 16 mm Hg means the ventricle is emptying long before it could open the aortic valve at about 80 mm Hg, so the exit is backward into the left atrium.

· Total stroke volume is large while forward output is low, which is the pattern of mitral regurgitation.

Distinctions that matter

· Losing volume during what should be isovolumetric contraction localises the leak to the mitral valve; gaining volume during what should be isovolumetric relaxation localises it to the aortic valve.

Common misconceptions

· Taking a wide loop as evidence of good forward output. The width here is largely regurgitant volume.

Where it is taught: Section C03 of the CV-4 Heart Valves and the Cardiac Cycle notes.

NPLEX-P1-CARD-04-01-C04-BD1

Brain dump 1 · foundational retrieval: Draw the jugular venous pulse against the heart sounds

A competent response contains

· a wave: right atrial contraction, after the P wave and just before S1.

· c wave: the closed tricuspid valve bulging into the atrium during right ventricular isovolumetric contraction, just after S1, with some transmitted carotid pulsation.

· x descent: the systolic fall in atrial pressure from atrial relaxation and the downward pull of the tricuspid ring as the ventricle ejects.

· v wave: atrial filling from the venae cavae against the closed tricuspid valve, peaking near S2.

· y descent: atrial emptying when the tricuspid valve opens, during rapid filling in early diastole.

· The right internal jugular vein runs nearly straight through the brachiocephalic vein and superior vena cava into the right atrium, so right atrial pressure changes appear in the neck.

· Differences from the carotid: the jugular pulse is not palpable, usually shows two peaks per cycle, falls with inspiration, and changes with position.

· The sternal angle lies about 5 cm above the right atrium, so with the patient at about 45 degrees a pulsation more than about 3 cm above the sternal angle suggests raised right atrial pressure.

· The left atrial or wedge tracing has the same waves.

Distinctions that matter

· The a wave is produced by the atrium contracting and the v wave by the atrium filling, so one precedes S1 and the other peaks near S2.

· The c wave comes from the valve bulging back into the atrium, not from blood crossing the valve.

Common misconceptions

· Calling the v wave right ventricular contraction. It is passive atrial filling against a closed valve.

· Timing the y descent in systole. The tricuspid valve opens in early diastole, which is when the atrium empties.

Where it is taught: Section C04, first two paragraphs and the jugular wave table of the CV-4 Heart Valves and the Cardiac Cycle notes.

NPLEX-P1-CARD-04-01-C04-BD2

Brain dump 2 · integration and application: Read four abnormal neck tracings

A competent response contains

· Patient 1: intermittent cannon a waves from complete heart block; the atria and ventricles beat independently, so on some beats the atrium contracts against a closed tricuspid valve and the contraction is forced backward into the neck, and because the rhythms are unrelated the coincidence happens only on some beats.

· Patient 2: regular cannon a waves, because atria and ventricles activate together on every beat, as in some junctional rhythms and in AV nodal reentrant tachycardia.

· Patient 3: absent a wave in atrial fibrillation, because there is no organised atrial contraction.

· Patient 4: tricuspid regurgitation; the atrium receives blood ejected backward from the right ventricle, so pressure rises throughout systole, the x descent is lost, the c and v waves merge into a large systolic c-v wave, the y descent is steep, and systolic hepatic pulsation may be felt.

· The wedge tracing in acute mitral regurgitation shows the same large v wave on the left side.

· A large a wave that is not a cannon wave means resistance to atrial emptying through an open valve, as in tricuspid stenosis or a stiff right ventricle.

· Ventricular tachycardia with AV dissociation and ventricular pacing also produce irregular cannon a waves.

Distinctions that matter

· Cannon a waves mean the atrium is contracting against a closed valve, while large a waves mean it is straining to empty through an open one.

· Regular against irregular cannon waves follows from whether the atria and ventricles are linked on every beat or beating independently.

Common misconceptions

· Reading any large neck pulsation as tricuspid regurgitation. Timing separates them: an a wave is presystolic and the c-v wave of regurgitation fills systole.

· Expecting cannon a waves in atrial fibrillation. Without organised atrial contraction there is no a wave at all.

Where it is taught: Section C04, abnormal wave paragraphs, the jugular wave table, and the stop and predict box of the CV-4 Heart Valves and the Cardiac Cycle notes.

NPLEX-P1-CARD-04-01-C04-Q02

Board-style question: Intermittent giant neck pulsations

Correct answer: A. Right atrial contraction against a closed tricuspid valve on beats where the timing coincides

Why it is right: In complete heart block the atria and ventricles beat independently, so on the beats where atrial contraction lands during ventricular systole the atrium contracts against a closed tricuspid valve and the pressure is forced back up the jugular vein as a cannon a wave, which appears only on those beats.

B is wrong: Tricuspid regurgitation raises atrial pressure through the whole of systole on every beat, giving a large c-v wave with a lost x descent rather than occasional pulsations.

C is wrong: Tricuspid stenosis obstructs emptying through an open valve, producing a large a wave on every beat with a slow y descent, not an intermittent giant wave.

D is wrong: A transmitted carotid pulsation would be palpable and would occur with every ejection, and the jugular pulse is specifically not palpable.

A competent response contains

· P waves at 92/min and QRS complexes at 40/min with no fixed relationship are complete heart block with AV dissociation.

· When a P wave falls during ventricular systole the right atrium contracts against a closed tricuspid valve, which forces the contraction backward into the neck as a cannon a wave.

· The pulsations are intermittent because the atrial and ventricular rhythms are unrelated, so the timing coincides only on some beats.

Distinctions that matter

· Intermittent cannon waves point to AV dissociation, while cannon waves on every beat point to atria and ventricles activating together.

Common misconceptions

· Reading a large venous pulsation as a sign of valve disease without timing it against the cardiac cycle.

Where it is taught: Section C04 of the CV-4 Heart Valves and the Cardiac Cycle notes.

NPLEX-P1-CARD-04-01-C05-BD1

Brain dump 1 · foundational retrieval: Build the S2 splitting table from memory

A competent response contains

· A2 is the aortic component, loud and heard widely; P2 is the pulmonic component, soft and best heard at the left 2nd intercostal space, which is where splitting is assessed; A2 normally comes first and a split is audible when the components are more than about 0.03 s apart.

· Physiologic: A2 then P2, single or very narrow in expiration, split by about 0.04 s on inspiration, because inspiration delays P2 and slightly advances A2; normal, especially in the young.

· Wide: A2 then P2, split in expiration and wider on inspiration, because P2 is delayed throughout or A2 is early; right bundle branch block, pulmonic stenosis, and severe mitral regurgitation through an early A2.

· Fixed: A2 then P2, a wide split that does not change with breathing, because the right ventricular volume load stays constant as shunt changes offset changes in venous return; atrial septal defect.

· Paradoxical: P2 then A2, split in expiration and single or narrower on inspiration, because A2 is delayed past P2; left bundle branch block, severe aortic stenosis, right ventricular pacing, and obstructive hypertrophic cardiomyopathy.

· The two bedside questions: is a split heard in expiration, and does inspiration widen it, leave it unchanged, or narrow it.

· Physiologic splitting is easiest to hear in children and young adults.

Distinctions that matter

· Wide and fixed splitting both give a split in expiration; what separates them is whether inspiration widens it further.

· Paradoxical splitting is the only pattern in which the components are reversed, and it is also the only one that narrows on inspiration.

Common misconceptions

· Listening for splitting at the apex. P2 is soft and is assessed at the left 2nd intercostal space.

· Treating any wide split as pathological without timing it to breathing. A physiologic split widens on inspiration and closes in expiration.

Where it is taught: Section C05, opening paragraph and the S2 splitting table of the CV-4 Heart Valves and the Cardiac Cycle notes.

NPLEX-P1-CARD-04-01-C05-BD2

Brain dump 2 · integration and application: Four patients with a split S2

A competent response contains

· Patient 1: wide splitting from right bundle branch block; the right ventricle is activated late so P2 is delayed throughout, and the respiratory mechanism still works, which is why inspiration widens it further.

· Patient 2: fixed splitting from an atrial septal defect; the left to right shunt volume loads the right ventricle and distends the pulmonary bed, delaying P2, and because the atria act as one chamber the extra systemic return of inspiration is offset by a smaller shunt, with the reverse in expiration, so right ventricular volume and P2 timing stay nearly constant.

· Patient 3: paradoxical splitting from severe aortic stenosis; A2 is delayed mechanically past P2, so the split is heard in expiration and the normal inspiratory delay of P2 moves it toward the late A2 and the sound becomes single.

· Patient 4: wide splitting from severe mitral regurgitation; A2 is early rather than P2 late, because the regurgitant orifice gives the left ventricle a second outlet so ejection finishes sooner, and the respiratory mechanism still works.

· Electrical delay of a component comes from late activation of that ventricle, as in right bundle branch block for P2 and left bundle branch block or right ventricular pacing for A2.

· Mechanical delay comes from prolonged ejection, as in pulmonic stenosis for P2 and severe aortic stenosis or obstructive hypertrophic cardiomyopathy for A2.

Distinctions that matter

· A wide split can come from a late P2 or an early A2, so name which component has moved rather than only the width.

· Fixed splitting is not simply a very wide split: the respiratory variation is abolished, which is what the shunt explains.

Common misconceptions

· Expecting inspiration to widen every split. It narrows a paradoxical split and leaves a fixed one unchanged.

· Assuming severe mitral regurgitation delays P2. It moves A2 earlier, and the width comes from that.

Where it is taught: Section C05, abnormal splitting paragraphs and the S2 splitting table of the CV-4 Heart Valves and the Cardiac Cycle notes.

NPLEX-P1-CARD-04-01-C05-RT01

Retrieval exercise: Rebuild the sequence that widens the split on inspiration

A competent response contains

· Step 1: inspiration makes intrathoracic pressure more negative.

· Step 2: systemic venous return to the right heart rises.

· Step 3: right ventricular ejection lasts longer.

· Step 4: pulmonary vascular impedance falls, which prolongs forward flow.

· Step 5: P2 is delayed.

· Step 6: blood pools in the lungs, left ventricular filling dips, and A2 comes slightly earlier.

· Step 7: the split widens.

· Steps 1 to 5 act on the right heart; step 6 acts on the left.

· The expanded pulmonary vessels briefly hold more blood, so less reaches the left ventricle, its stroke volume dips, ejection is shorter, and A2 moves earlier.

· The split goes from single or very narrow in expiration to about 0.04 s on inspiration, with about 0.03 s needed for a split to be audible.

· It is heard most easily in children and young adults.

Distinctions that matter

· Breathing changes right heart filling more than left, which is why the split widens rather than simply shifting both components.

· Two separate changes contribute: P2 moves later and A2 moves slightly earlier.

Common misconceptions

· Explaining the widening by venous return alone. Falling pulmonary impedance also prolongs right ventricular ejection.

· Expecting left ventricular filling to rise on inspiration along with the right. It dips, because blood is briefly held in the expanded pulmonary vessels.

Where it is taught: Section C05, physiologic splitting paragraph and flow diagram of the CV-4 Heart Valves and the Cardiac Cycle notes.

NPLEX-P1-CARD-04-01-C06-BD1

Brain dump 1 · foundational retrieval: S3 and S4 side by side

A competent response contains

· S3: early diastole, about 0.12 to 0.16 s after S2, during rapid filling; a large fast inflow is abruptly decelerated as the ventricle reaches its limit of expansion, setting walls and blood vibrating.

· S4: late diastole just before S1, with atrial contraction; atrial ejection into a stiff ventricle causes a sharp pressure rise and a low frequency vibration.

· Both are low pitched and made by the ventricle and its blood, not by valves.

· Left-sided sounds are heard best with the bell at the apex in the left lateral position; right-sided sounds at the left lower sternal border, and they are louder on inspiration because inspiration increases right heart filling.

· S3 is often normal in children, young adults up to about age 40, trained athletes, and pregnancy, where a compliant ventricle fills briskly.

· In older adults S3 signals volume overload: a dilated failing ventricle with high atrial pressure, regurgitant volume returning in mitral or aortic regurgitation, a left to right shunt, or a high output state.

· S4 indicates reduced compliance: hypertrophy from hypertension, aortic stenosis, or hypertrophic cardiomyopathy, and acute ischemia, which impairs relaxation; it may occur with age-related stiffening but is abnormal in the young.

· S4 is absent in atrial fibrillation because it requires organised atrial contraction; S3 can persist in atrial fibrillation because it depends on passive rapid filling.

· Significant mitral stenosis prevents the brisk early inflow that makes an S3, so no S3 occurs.

Distinctions that matter

· S3 is a volume sound in early diastole and S4 is a stiffness sound in late diastole, which is why atrial fibrillation abolishes one and not the other.

· An S3 can be a normal finding in the young, whereas an S4 in a young person is abnormal.

Common misconceptions

· Attributing S3 or S4 to a valve snapping. Both come from the ventricular wall and the blood inside it.

· Expecting an S3 in severe mitral stenosis because atrial pressure is high. The narrowed valve prevents the rapid inflow the sound depends on.

Where it is taught: Section C06, S3 and S4 paragraphs and the stop and predict box of the CV-4 Heart Valves and the Cardiac Cycle notes.

NPLEX-P1-CARD-04-01-C06-BD2

Brain dump 2 · integration and application: Decide which murmur is just flow

A competent response contains

· A murmur is turbulent flow; turbulence appears when the Reynolds number exceeds a critical value, and that number rises with flow velocity and vessel diameter and falls with viscosity.

· Pregnancy: blood volume rises about 40% to 50% and cardiac output about 30% to 50%, so velocity rises; the murmur goes after delivery.

· Anemia: hypoxic vasodilation raises cardiac output and low red cell mass lowers viscosity, so both terms push toward turbulence; the murmur goes when the anemia is corrected.

· A six-year-old: high output relative to body size with a thin chest wall that transmits sound, so a normal valve is audible; the murmur goes as the child grows.

· Fever at 39.4 degrees C (103 degrees F): raised cardiac output and velocity; the murmur goes when the fever settles. Thyrotoxicosis and exercise act the same way.

· Functional murmurs are soft, grade 1 or 2 with no thrill, early to mid systolic ejection murmurs at the base or left sternal border, with normal S2 splitting and no extra sounds.

· Findings that argue for structural disease: a diastolic murmur, a holosystolic murmur, a thrill, or an abnormal S2.

· The venous hum is the innocent continuous murmur, heard above the clavicle, and it disappears when the child lies down or the jugular vein is compressed.

· Common innocent murmurs in children: a vibratory musical early systolic murmur at the left lower sternal border or apex known as the Still murmur, a soft pulmonary flow murmur at the left 2nd intercostal space, and the venous hum above the clavicle.

Distinctions that matter

· High flow and low viscosity both raise the Reynolds number, so a flow murmur can arise from more output, thinner blood, or both.

· Timing is the strongest structural clue: a diastolic or holosystolic murmur is never functional, while the venous hum is continuous and still innocent.

Common misconceptions

· Reading any murmur in a child or a pregnant woman as innocent. Grade, timing, S2, and extra sounds still have to be checked.

· Assuming a functional murmur means a mild structural lesion. The valve is normal, and removing the cause removes the murmur.

Where it is taught: Section C06, functional flow murmur paragraphs of the CV-4 Heart Valves and the Cardiac Cycle notes.

NPLEX-P1-CARD-04-01-C07-BD1

Brain dump 1 · foundational retrieval: Build the murmur timing table from memory

A competent response contains

· Rule: stenosis is heard when the valve should be open and regurgitation when it should be closed, so aortic and pulmonic stenosis are systolic while their regurgitations are diastolic, and mitral and tricuspid regurgitation are systolic while their stenoses are diastolic.

· Grading runs 1 for faint to 6 for audible with the stethoscope off the chest, and a palpable thrill means grade 4 or higher.

· Ejection crescendo-decrescendo: aortic stenosis at the right 2nd intercostal space with carotid radiation and a soft A2; pulmonic stenosis at the left 2nd intercostal space with a wide split S2; hypertrophic cardiomyopathy at the left lower sternal border with no carotid radiation.

· Holosystolic: mitral regurgitation at the apex radiating to the left axilla; tricuspid regurgitation at the left lower sternal border and louder on inspiration; ventricular septal defect at the left lower sternal border and harsh.

· Click then late systolic: mitral valve prolapse at the apex, with the click earlier when the left ventricle is smaller.

· Early diastolic decrescendo: aortic regurgitation at the left sternal border in the 3rd to 4th space with a wide pulse pressure, best heard sitting forward in held expiration.

· Opening snap then mid-diastolic rumble: mitral stenosis at the apex in the left lateral position with the bell, with presystolic accentuation.

· Mid-diastolic rumble: tricuspid stenosis at the left lower sternal border, louder on inspiration.

· Continuous peaking at S2: patent ductus arteriosus at the left infraclavicular area, machinery-like, because aortic pressure exceeds pulmonary pressure throughout the cycle.

· An ejection murmur starts after S1 because flow across the outflow valve begins only after isovolumetric contraction, and it ends before S2.

· A holosystolic murmur runs from S1 to S2 because blood escapes into a lower pressure chamber from the onset of contraction; a small ventricular septal defect may be loud because its jet is fast.

· In mitral valve prolapse the cusps hold until the ventricle becomes small enough in mid systole, so a midsystolic click precedes a late systolic murmur.

· As aortic stenosis worsens the murmur peaks later and A2 softens; in mitral stenosis a shorter S2 to opening snap interval means higher atrial pressure and more severe disease.

Distinctions that matter

· Aortic stenosis and hypertrophic cardiomyopathy are both ejection murmurs, and carotid radiation separates them.

· Timing within systole carries the diagnosis: after S1 and ending before S2 means an outflow obstruction, S1 to S2 throughout means a leak into a lower pressure chamber, and late with a click means prolapse.

Common misconceptions

· Taking a louder murmur as a more severe lesion. Loudness depends on flow as well as severity, so a severely stenotic valve may soften as output falls and a small septal defect may be louder than a large one.

· Expecting A2 to be loud in severe aortic stenosis. It softens as the valve becomes immobile.

Where it is taught: Section C07, systolic, diastolic, and continuous murmur paragraphs and the murmur timing table of the CV-4 Heart Valves and the Cardiac Cycle notes.

NPLEX-P1-CARD-04-01-C07-BD2

Brain dump 2 · integration and application: Separate two systolic murmurs with the bedside maneuvers

A competent response contains

· Inspiration increases right heart return, so right-sided murmurs get louder while hypertrophic cardiomyopathy and mitral valve prolapse change little.

· Squatting raises both preload and afterload, so most murmurs get louder, hypertrophic cardiomyopathy gets softer, and the prolapse click moves later.

· Standing from a squat lowers preload, so most murmurs get softer, hypertrophic cardiomyopathy gets louder, and the prolapse click moves earlier.

· Handgrip raises afterload, so mitral regurgitation, aortic regurgitation, and ventricular septal defect get louder while the aortic stenosis gradient and murmur fall; hypertrophic cardiomyopathy gets softer and the prolapse click moves later.

· Valsalva strain lowers preload, so most murmurs get softer, hypertrophic cardiomyopathy gets louder, and the prolapse click moves earlier.

· More preload means a larger stroke volume across the lesion and a louder murmur; more afterload increases the backward gradient in a regurgitant lesion and makes those murmurs louder.

· The two conditions that grow louder on standing are obstructive hypertrophic cardiomyopathy and mitral valve prolapse, and both worsen as the left ventricular cavity shrinks.

· In hypertrophic cardiomyopathy a smaller cavity brings the hypertrophied septum and the anterior mitral cusp closer together and worsens outflow obstruction; in prolapse a smaller cavity lets the redundant cusps prolapse earlier in contraction.

· In aortic stenosis the obstruction is fixed, so a lower preload only reduces flow across it and softens the murmur, and handgrip reduces the gradient, which also softens it.

Distinctions that matter

· Most murmurs follow the volume crossing the lesion, while hypertrophic cardiomyopathy and mitral valve prolapse follow cavity size and therefore move the opposite way with preload.

· Preload and afterload are separate levers: squatting raises both, Valsalva and standing drop preload, and handgrip raises afterload alone.

Common misconceptions

· Expecting handgrip to make every murmur louder. It increases regurgitant murmurs and softens the aortic stenosis murmur.

· Treating the prolapse click and murmur as moving together in loudness alone. The click shifts in timing, and the murmur lengthens as the click moves earlier.

Where it is taught: Section C07, dynamic maneuvers paragraph and the maneuver table of the CV-4 Heart Valves and the Cardiac Cycle notes.

NPLEX-P1-CARD-04-01-C07-PR01

Prediction exercise: Predict what standing does to a click and a late systolic murmur

A competent response contains

· Venous return falls on standing, so left ventricular end-diastolic volume falls.

· The cavity is smaller at the start of systole.

· The redundant cusps reach the point of giving way earlier in contraction, because a smaller cavity lets them prolapse sooner.

· The click moves earlier in systole.

· The murmur becomes longer, since it runs from the click to S2, and it is a late systolic murmur that now begins sooner.

· Squatting raises preload and afterload, so the cavity is larger, the click moves later, and the murmur shortens; handgrip raises afterload, which also moves the click later.

Distinctions that matter

· The click reports the moment the cusps give way and the murmur reports the leak that follows, so the two findings move together in timing.

· Mitral valve prolapse moves opposite to most murmurs with preload, in the same direction as hypertrophic cardiomyopathy.

Common misconceptions

· Expecting a lower preload to soften this murmur as it does most murmurs. A smaller cavity makes the prolapse worse.

· Reading an earlier click as a shorter murmur. An earlier click leaves more of systole for the murmur to fill.

Where it is taught: Section C07, dynamic maneuvers paragraph, the maneuver table, and the stop and predict box of the CV-4 Heart Valves and the Cardiac Cycle notes.